
◆ Instruction (Statements) in C

Instruction matlab ek complete command jo computer ko batata hai kya karna hai. Har instruction ke end mein **semicolon (;)** lagta hai.

👉 Example:

```
int a = 5;
```

Yeh ek instruction hai.

📌 Types of Instructions

1. Type Declaration Instruction

Variables declare karne ke liye use hota hai.

👉 Example:

```
int a;  
float b;  
char ch;
```

2. Arithmetic Instruction

Math calculations ke liye use hota hai.

👉 Example:

```
int sum = a + b;  
int mul = a * b;
```

3. Control Instructions

Program ka flow control karta hai.

✓ **Decision Making (if-else)**

```
if (a > b) {  
    printf("a bada hai");  
} else {  
    printf("b bada hai");  
}
```

✓ **Loop (repeat karna)**

```
for(int i = 0; i < 5; i++) {  
    printf("%d", i);  
}
```

✓ **Switch Case**

```
switch(choice) {  
    case 1: printf("One"); break;  
}
```

4. Input/Output Instruction

User se input lena aur output dikhana.

```
scanf("%d", &a);  
printf("%d", a);
```

◆ Operators in C

Operators symbols hote hain jo operation perform karte hain (jaise +, -, *, etc.)

Types of Operators

1. Arithmetic Operators

Math ke liye:

- + Addition
- - Subtraction
- * Multiplication
- / Division
- % Modulus

👉 Example:

```
int c = a + b;
```

2. Relational Operators

Compare karne ke liye:

- == equal
- != not equal
- > greater
- < less

👉 Example:

```
if (a > b)
```

3. Logical Operators

Conditions combine karne ke liye:

- && (AND)
- || (OR)
- ! (NOT)

👉 Example:

```
if (a > 0 && b > 0)
```

4. Assignment Operators

Value assign karne ke liye:

- `=`
- `+=`, `-=`, `*=`

👉 Example:

```
a += 5; // a = a + 5
```

5. Increment/Decrement

- `++` (increase)
- `--` (decrease)

👉 Example:

```
i++;
```

6. Ternary Operator

Shortcut of if-else:

```
int max = (a > b) ? a : b;
```

Easy Difference

- **Instruction** → pura statement
👉 `a = b + c;`
- **Operator** → symbol jo kaam karta hai
👉 `+`, `=`, `>`